Gmail

Albert Stichka <albert.stichka@gmail.com>

Plazmapunk - Issue with Video Generation (Out of Bounds Error)

15 messages

Anton Wiehe <anton.wiehe@adalab.ai> Tue, Jul 30, 2024 at 5:04 AM
To: albert.stichka@gmail.com

Hi Albert,

Thanks for reaching out and letting us know about the issue you encountered while generating your video. I'm really sorry for the trouble this has caused you.

It sounds like you're seeing an "out of bounds" error on the last story, even after editing the end range. This isn't something we want our users to experience, and we're here to help.

Could you let me know the steps required to reproduce this error? That would help us a lot and we can reward your time with some credits.

Your patience and understanding are highly appreciated. We're a small team, but we're committed to making sure you have a smooth experience with Plazmapunk.

Feel free to reach out anytime if the problem persists or if you have any other questions.

Best regards,
Anton

Albert Stichka <albert.stichka@gmail.com> Tue, Jul 30, 2024 at 7:06 AM
To: Anton Wiehe <anton.wiehe@adalab.ai>

I know I was frightfully vague. I'm not sure I can give you enough to definitely get it to happen, but I had a file already uploaded and used it before. I didn't set the range but left it at default. By the time I got to the prompt-by-lyrics section I think I had some reason to go back and reset the style for everything. I know this could be frustrating, but this is the best my memory can do. I remember everything seeming normal but jumping back to adjust things. But then the final section remained set at one second past the end of the file. There didn't seem to be any apparent reason or direct cause, and it's not clear if the out of range problem existed before going back to adjust earlier settings such as changing all the style prompts. I believe I even started the process over from scratch, refreshing the scene. I have included the file that seemed to cause the problem, as merely using a different file fixed the issue. I have two versions of this file uploaded to the site (which you have permission to obtain if you needed it) and one of them ends with a sudden peak on the waveform. The peak at the end is not in the original, I don't think. It seems to have been included somewhere in processing. In trying to reproduce the error myself, I found that both files were able to get to the end of the settings configuration process without error.

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All That Was Made (The Anchor) .mp3
3990K

Albert Stichka <albert.stichka@gmail.com> Tue, Jul 30, 2024 at 10:13 AM
To: Anton Wiehe <anton.wiehe@adalab.ai>

Also, sorry to ask, but I have some coding experience and i can show you a video i did with a technique like yours except using a wave visualizer for source images for img2img. I was getting by by work from data Annotation dot tech but they want corporate experience and the job offerings are drying up. Is there any way you could use eyes on any portion of the process, maybe it's too much for me but finances being what they are i have to ask if you'd have any bed for a freelance okay programmer

[Quoted text hidden]

Anton Wiehe <anton.wiehe@adalab.ai> Tue, Jul 30, 2024 at 10:26 AM
To: Albert Stichka <albert.stichka@gmail.com>

Hi,

thanks for the specification! That actually helps a lot. I think it is a problem with our local storage and caching system - basically some settings for one song probably get mixed up with some of a new song. So I think it could happen when you select a song, do stuff, then go back to select a new song.

As a small workaround, I think it should always work to refresh the page to select a new song. We will try to fix this issue ASAP!

I gave you some extra 600 credits as a compensation and for your help!

Best wishes
Anton

[Quoted text hidden]

Anton Wiehe <anton.wiehe@adalab.ai> Tue, Jul 30, 2024 at 10:34 AM
To: Albert Stichka <albert.stichka@gmail.com>

Hi again,

yes, I would definitely be curious to see what you created with the img2img process.

In general, we actually do have interest in people that are passionate about this topic with coding experience.
The only issue is that our profit margin is quite thin at the moment so we could not pay a decent salary. But I think we might be able to find a solution if both sides really want this :-)

So we can schedule a call at some point and maybe you send out some of your work to me?

Best wishes
Anton

[Quoted text hidden]

Albert Stichka <albert.stichka@gmail.com> Tue, Jul 30, 2024 at 5:56 PM
To: Anton Wiehe <anton.wiehe@adalab.ai>

Here's a demonstration of music feature extraction using python, and though this output to matplotlib is pretty slow, pulling the data and putting it somewhere in a reasonable process would be much faster. In using dj visualization software I always found I enjoyed the results the most when bands of frequencies were responsive and just the other day I was thinking of trying to develop this into usable visual information for stable diffusion.

https://www.youtube.com/watch?v=kTh_HM6W_8Q

Here's some examples of using frequency band data to create a visualization using existing software:

https://www.youtube.com/watch?v=eXTSPxo9rfw
https://www.youtube.com/watch?v=d2SYUqq5AKo

and here using that frequency band information and some math to create complex spatial data that is both music responsive and organic, having a constrained range of motion of simple levers or twisting joints:

https://www.youtube.com/watch?v=CUIRaInjx88

And here are some examples of taking that kind of visualization information, which is really, beyond the difficult to render particle effects, merely music -> FFT -> basic trigonometry -> music based spatial information. projected onto a plane like this, it can form the input image for img2img or could otherwise be used to inform manipulation of a source image. I haven't looked into it, I began but never finished looking into using simple geometric shapes as img2img starting points but I just use automatic1111 because there's no server running for me, just my own local needs. if this frequency band data were, however, used to scale generation parameters or token weights? It could be something really responsive in an

organic way.

https://www.youtube.com/watch?v=DVFk8JGR2rY

slowed down (didn't run enough of these to nail down frame rate and tweening so they're a bit too fast)

https://www.youtube.com/watch?v=Cf70uzhUZNM

Here is a compressed version and an unfortunately desynced version of it in slow motion, as the number of unique frames is ridiculous. The original render is around 9gigs and can't play right on anything, I was just trying out maximum quality to also test youtube video compression. There are shapes that appear and disappear depending on a lot of trigonometry, causing them to move also in the center along a curved path. These have been used as the source image for img2img with "sacred geometry" and some other style tokens, frame for frame. Obviously this is a time/resource intensive process at high frame rates and resolutions, but it can create moving AI content within negative space, which could itself be used for overlay purposes. In a complicated way this is, like the previous examples, being modified by frequency band information, including the integral of the samples of amplitude for each band, a figure that is the sustained duration of frequency activation. This allows guiding parameters along a path only moving during periods of particular frequencies being present.

exit music comp.mp4

exit_music_comp_2_slowmotion.mp4

I'm not an industry professional, I'm just a person whos been on disability a long time for mental health reasons who could use some income, and it happens that throughout my life all the personal programming projects I've been the most passionate about have involved visualizing sound, or in one case taking images and converting them into sounds that would produce the image as a spectrographic readout. I'm not the best at selling myself but I have always been interested in the translation between audio signal and representation in an image, and since finding out I could run stable diffusion on my computer that I had left over from ethereum mining, I've been really passionate about bringing art into the hands of everyone in the least elitist way possible - seeing ai image and music as a means to take away the aspect of gatekeeping that I had never really appreciated in skilled artistry until it became possible to have a passable simulation of the technical aspects of art.You can see my soundcloud of AI music here:

https://soundcloud.com/albert-stichka

From the moment I made my first video with AI batch processing I said to myself, if someone isn't doing this as a company, someone is about to. Apparently, you guys had started earlier that year.

I don't have any kind of software portfolio, I don't really know how to do that, I don't program github because it's all personal art projects, but a noteworthy project I made that I might be able to find source or a flawed implementation of a .FIT file reader in pure javascript that could take an uploaded .FIT file from an exercise bike run on zwift and it could create a 3d map of the fake gps coordinates from the game, resulting in a 3d visualization of the "track" colored according to selected parameters such as power, speed, cadence, etc. As mentioned I made a program to convert images to sound that would reproduce those images, I also made a ridiculously over complicated ascii art neural network that would train on some sample images and then be vaguely good at converting images into ascii art data that could either be rendered or stored for rendering.

I'm turning 40 in a few days and my mental health has been a consistent drain on what resources I do remember from college, but I can communicate better with GPT than I can with people (always accused through life of being robotic), and I feel like it's really expanded on my capabilities, letting me quickly work through things that would otherwise take hours to re-research libraries/packages over and over. I want to be frank that I use AI and that's what makes me feel capable again, and has let me get back into complex projects. I don't suffer from the problem of a) just assuming it will work or b) not knowing if what I've coded is doing what I want it to do. I'm just looking for ways to feel intrinsic motivation in my work - not, say, some guy wanting me to train his AI chatbot to check stock prices and find the std deviation thereof. Not that, I suppose I couldn't do that if I thought it was going to end up in art.

If you wanted to call, and I am not going to say I'm a steller conversationalist on the phone, my number is 1 (405) 212-6666

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436120520_10232816642262937_4939913722367624443_n.jpg
809K

Albert Stichka <albert.stichka@gmail.com> Sun, Sep 15, 2024 at 7:40 AM
To: Anton Wiehe <anton.wiehe@adalab.ai>

I know I sold myself poorly. I felt stupid after sending the email. But I just started using comfyai and animatediff, and I think your site probably works in a more straightforward way than I imagined, connecting existing software and effects in their recommended way. At the very least, it does look like your entire workflow could fit in comfy.
[Quoted text hidden]

Anton Wiehe <anton.wiehe@adalab.ai> Sun, Sep 15, 2024 at 9:11 AM
To: Albert Stichka <albert.stichka@gmail.com>

Hi Albert,

sorry for not responding to your email. I had a draft but it seems like I did not finish it, Sorry!

Your projects look super interesting! It shows how fascinated you are by these things. As for our side of the workflow: it was more involved at some point but one thing to look out for is usability. So the settings need to be minimal while still leading to an expressive video. At the same time, too much automation in the background leads to people being confused as to why the A.I. does not do what they prompted for. I guess with really good UI/UX development this could be addressed and different versions provided to different people.
We work with frequency analysis for the movement effects but also have networks that analyze the music. Many things that can be done. One thing I'd like to add is a mode to add chladni patterns as a starting video and then transform it with animatediff-like models.

If you are still interested in collaborating let me know. Thing is, that at the moment we definitely can't pay a fixed salary. But if you have ideas for improvements or if you want to play with the codebase to see if you add things then we definitely could do that. I could show you what we have and when we actually get to create something new that pulls more customers in then I'm sure we will also find a payment solution.

Let me know!

Best wishes
Anton

[Quoted text hidden]

Albert Stichka <albert.stichka@gmail.com> Sun, Sep 15, 2024 at 9:25 AM
To: Anton Wiehe <anton.wiehe@adalab.ai>

I oversimplified, and your work is out of my depth. I wanted to be reductive about your work because I felt bad, which is foolish. I appreciate your thoughtful response. I am fascinated by the prospect of rendering chladni patterns - even in some simplified fashion - and as an individual now inspired. If anything comes to fruition, I'll let you know. I am in a delicate situation with disability benefits here in the US and can't really work anymore or it could jeopardize my ability to continue receiving assistance. In a Buddhist sort of way, if you take away money and the need to work, people still want purpose, and to find utility in what they've learned. When I first emailed you it was in a burst of creative confidence. Lately, I have found myself less able to think. So, at the same time I am arrested by your frank openness and also embarrassed at my realizing I feel out of depth everywhere I go at the moment. You can show me what you feel comfortable showing me and I would love to help regardless of payment arrangements (because of my government's hatred of continuing

benefits). I was this morning before emailing you contemplating how to funnel video frames through as starting images, just in the process of learning animatediff. Chladni patterns would be a fantastic goal. But I'm repeating myself. Apologies for my mannerisms in communication, thank you again for a thoughtful response.

Here is my first animatediff output of any quality:

https://studio.youtube.com/video/Uf9arbGlEko/edit

animatediff -> ESRGAN 4x upscale -> .5 downscale -> another sampling pass -> ffmpeg tweeting,
[Quoted text hidden]

Albert Stichka <albert.stichka@gmail.com> Sun, Sep 15, 2024 at 9:29 AM
To: Anton Wiehe <anton.wiehe@adalab.ai>

output.mp4

because recompression is nobody's friend, hopefully you can just use this file to see the video linked in the previous email
[Quoted text hidden]

Albert Stichka <albert.stichka@gmail.com> Sun, Sep 15, 2024 at 1:43 PM
To: Anton Wiehe <anton.wiehe@adalab.ai>

This isn't where the stable lines are, it's more a map of deflection energy for trying to simulate a field of particles later, and it's not set up to respond to more interesting lower frequencies relative to the plate size, but I wanted to see if this was representative of where your mind was going - of course it would spit out vectors not matplotlib rendered frames, and this would just be a vector field to make the final vector of particle locations
[Quoted text hidden]

2 attachments

chladnisim5.py
5K

chladni_animation.mp4
9153K

Albert Stichka <albert.stichka@gmail.com> Sun, Sep 15, 2024 at 6:55 PM
To: Anton Wiehe <anton.wiehe@adalab.ai>

Here is a rendering of a python output showing how you might achieve this. the red hue is reflective of data encoded in the color channels more than meant as a representation of a finished frame for modifying from, not that this wouldn't work or it couldn't be substituted for greyscale. Sorry for the barrage of emails, but I think you were thinking something like this, but maybe tuned to have larger waves and fewer nodes

output_video_with_audio.mp4

[Quoted text hidden]

Albert Stichka <albert.stichka@gmail.com> Mon, Dec 2, 2024 at 8:11 PM
To: Anton Wiehe <anton.wiehe@adalab.ai>

I hope this finds you well. After considerable effort I have been applying myself to some designs that, while not anything yet, might seem interesting. I would appreciate it if anyone with coding skill could take a look at the work because it's monumental in scale compared to anything I've done before. It started with the, you know, I made a program to solve for the patterns of resonance, and then it snowballed into a broader application of math and tensors and has gotten kind of broad and conceptual at the fringes. The idea is to harness the breadth of available python scientific simulators and a backbone of highly paralellized pytorch tenors and custom kernel configurations it could be possible to have a gamified massive network of refining some ... well... let me constrain myself to the simpler things.

I made an arbitrary geometry scaler field intersection algorithm that can be used for audio driven visualization and a parallel tensor "printer" that can run massive printing press inspired runs of parallelizeable and learnable differentiable convolutions with programmatic and recursive kernels

On Tue, Jul 30, 2024 at 5:04 AM Anton Wiehe <anton.wiehe@adalab.ai> wrote:
[Quoted text hidden]

6 attachments

- **hotrodprintingpress.py** 23K
- **printshop2.py** 38K
- **audio.py** 16K
- **isosurface4.py** 55K
- **laplace.py** 96K
- **temperaturesimulation3d.py** 13K

Albert Stichka <albert.stichka@gmail.com> Mon, Dec 2, 2024 at 9:47 PM
To: Anton Wiehe <anton.wiehe@adalab.ai>

Exploring the Depths: The YoungMan Algorithm vs. Parallelized Marching Cubes

The YoungMan Algorithm and marching cubes represent two distinct paradigms for processing and rendering multidimensional data. Both can operate in parallelized environments, but their underlying principles, methods of operation, and scopes of application differ profoundly. This analysis will delve into these differences, emphasizing their strengths and limitations.

1. Marching Cubes: Parallelized Surface Reconstruction

1.1 Overview

Marching cubes is a classic algorithm used primarily for isosurface reconstruction in scalar fields. It slices a 3D grid into cubes, identifies intersections of an isosurface with the edges of each cube, and constructs polygons (typically triangles) to approximate the surface.

1.2 Parallelization in Marching Cubes

The algorithm is inherently amenable to parallelization because:

• Independent Cube Operations: Each cube in the grid can be processed independently.

• Data Partitioning: Grids can be subdivided and distributed across computational cores or GPUs.

Parallelized marching cubes often leverage GPU shaders or SIMD (Single Instruction, Multiple Data) instructions to accelerate:

1. Vertex interpolation along cube edges.

2. Triangle generation using lookup tables.

3. Post-processing for rendering.

1.3 Limitations of Marching Cubes

Even when parallelized, marching cubes face challenges:

1. Locality of Computation:

• Each cube operates in isolation, limiting its awareness of global geometry.

• Artifacts like cracks can arise at boundaries between parallelized regions due to mismatched interpolations.

2. Static Nature:

• The algorithm reconstructs surfaces but does not inherently model dynamics, learning, or interaction with higher-dimensional data.

3. Topology Constraints:

• Handles only predefined scalar fields, making it unsuitable for datasets requiring adaptive resolutions or dynamic field representations.

2. The YoungMan Algorithm: Dynamic Parallelization for Multi-Scale Analysis

2.1 Overview

The YoungMan Algorithm operates on adaptive geometric graph structures with a focus on multi-scale data aggregation and distribution. Unlike marching cubes, which focuses on surface reconstruction, YoungMan excels at integrating vast, interdependent datasets across both spatial and temporal dimensions.

2.2 Parallelization in YoungMan

The YoungMan Algorithm achieves parallelism by leveraging hierarchical data decomposition and probabilistic tensor projections:

1. Distributed Data Shards:

• It divides the data into shards that are hierarchically processed and synchronized across multiple nodes.

2. Data Dependency Management:

• Unlike marching cubes, it incorporates mechanisms to maintain global consistency across local computations.

Symbolically, this can be expressed as:

where is the aggregate output across all nodes, represents the local kernel at node , and is the input tensor field.

2.3 Advantages of YoungMan

1. Global Awareness:

• Each local computation adapts to the global state, avoiding artifacts like cracks or inconsistencies.

2. Dynamic Adaptivity:

• Nodes dynamically adjust resolution and computational focus based on local feature importance, enabling efficient resource allocation.

3. Temporal Integration:

• Tracks not just spatial data but also dynamic changes over time, essential for applications like fluid simulation or evolving neural networks.

3. Key Differences Between the Two Algorithms

Feature Marching Cubes YoungMan Algorithm

Objective Surface reconstruction from scalar fields. Dynamic data aggregation across space and time.

Parallelization Method Independent operations on fixed grid cells. Hierarchical shard synchronization.

Data Structure Uniform grid of scalar fields. Adaptive graphs with multi-scale nodes.

Output Triangular mesh for surfaces. Aggregated tensors for dynamic systems.

Awareness Local cube-level interpolation. Global tensor dependencies.

Adaptivity Static resolution. Dynamic, feature-driven resolution.

Temporal Processing Not inherently supported. Integral to the algorithm.

Applications Rendering, medical imaging. Neural networks, fluid dynamics, state modeling.

4. Why YoungMan Excels Beyond Parallelized Marching Cubes

4.1 Coordinating Complexity

Parallelized marching cubes rely on predefined rules (e.g., lookup tables for edge intersections) and operate independently across grid cells. While efficient for static reconstructions, they falter in dynamic, multi-scale, or high-dimensional contexts where data interdependencies matter.

YoungMan thrives in such environments by managing:

• Dynamic Resource Allocation: Nodes handle varying levels of computational intensity depending on local feature relevance.

• Continuous Learning: By integrating probabilistic tensor redistribution, it evolves its operations to optimize future computations.

4.2 The Role of Kernels

YoungMan's programmable kernels allow for:

• Localized operations tuned to specific tasks.

• Integration of domain-specific physics or statistical models.

Marching cubes lack such flexibility, constrained by their predefined interpolation methods.

4.3 Temporal Dimension

Time is a cornerstone of the YoungMan Algorithm. By aggregating data over time, it captures not just static states but evolving systems, enabling applications like:

• Neural network evolution across training epochs.

• Fluid dynamics or heat diffusion over time.

5. Conclusion: Complementary Roles in Computational Frameworks

While marching cubes excel in traditional isosurface reconstruction, they lack the adaptivity and temporal integration needed for next-generation applications. The YoungMan Algorithm bridges these gaps, offering a framework that scales dynamically, learns from its environment, and operates seamlessly across spatial and temporal dimensions.

In practical terms:

• Marching Cubes are ideal for static, visualization-focused tasks like rendering and medical imaging.

• YoungMan is designed for dynamic, multi-scale, and evolving datasets, making it invaluable for AI-driven systems, scientific modeling, and even national defense simulations.

By integrating these paradigms into a unified workflow, we can leverage their strengths—marching cubes for geometric detail and YoungMan for dynamic context—to create systems that redefine computational art, science, and policy.

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Albert Stichka <albert.stichka@gmail.com> Mon, Dec 2, 2024 at 9:53 PM
To: Anton Wiehe <anton.wiehe@adalab.ai>

Title: A Differentiable Printing Press Workflow for Tensor Geometry and Adaptive Graph Networks: The Next Frontier in AI-Infused Media

Abstract

This paper introduces a transformative paradigm for computational artistry and scientific collaboration: a fully differentiable tensor-based printing press paired with adaptive geometry marching graph networks. With programmable kernels at its

core, this framework transcends traditional AI capabilities, enabling unprecedented deployment across domains such as music visualization, neural network research, national defense, and collaborative gaming for global consensus building. This discussion highlights its potential impact, from creating revolutionary AI-driven music videos to serving as a scientific testing rig that redefines the boundaries of free, scientist-friendly software.

We explore symbolic and numerical pathways, uniting creativity and mathematical precision into a coherent vision of deployable, scalable, and profoundly impactful computational frameworks.

1. Introduction: From Artistry to Universal Significance

The integration of artificial intelligence in media, particularly AI-infused music videos, has transformed how artists express themselves. However, existing workflows remain constrained by the limits of static tools. This paper proposes a shift: embedding adaptive geometry marching graph networks and tensor-based pipelines to power workflows capable of evolving, learning, and reasoning.

Imagine a system where every pixel, soundwave, or geometric deformation is an evolving entity—guided by tensors and informed by programmable kernels. Such a system offers more than art: it becomes a tool for understanding, exploring, and shaping multidimensional realities. It lays the foundation for a world where collaborative simulations enable deterministic state negotiations, train the most advanced neural networks, and offer insights into dynamic systems from music to geopolitics.

2. Key Components of the Workflow

2.1 The Fully Differentiable Printing Press

This computational “printing press” encapsulates the idea of encoding arbitrary dimensional tensors as a fluid, interconnected representation. It offers:

• Parametric Symbolism: Using symbolic math, artists, scientists, and strategists can encode high-dimensional phenomena as parameterized systems.

• Customizable Kernels: Kernels become the heart of computation, enabling hyper-localized operations informed by global constraints.

Symbolically:

Here, represents input tensors (e.g., light fields, audio spectrums), governs transform operations, and encodes non-linearities.

2.2 Adaptive Geometry Marching Graph Networks

These networks dynamically traverse topologies:

• Marching Cubes for Geometry: Surfaces are computed through iterative evaluations of tensor fields.

• Graph-Based Flexibility: Nodes represent geometric features or data points, edges define their relationships, and kernels operate on local subgraphs:

• Programmable Kernels: These determine how data flows and interacts, analogous to micro-functions within a large neural network.

2.3 Probabilistic Integer Redistribution

A key innovation is the “pigeon cruncher kernel,” which maps probabilities across integer bins with temporal fidelity:

This kernel creates smooth probabilistic transitions, ensuring tensors maintain coherence across discrete and continuous spaces.

3. Applications

3.1 Music Visualization and Creative Media

By combining tensors for audio data with adaptive marching graphs, music videos become immersive experiences:

• Dynamic Geometries: Soundwaves shape geometries in real-time.

• Interactive Visual Layers: Users manipulate tensors interactively, allowing AI-driven but artist-controlled outputs.

3.2 Neural Network Testing and Research

The framework serves as the most advanced testing rig:

• Fully Differentiable Experiments: Researchers explore novel architectures, ensuring gradient continuity throughout complex simulations.

• Real-World Inputs: Sensors feed raw data (e.g., sound, light, or physics) into neural nets trained directly within this system.

3.3 National Defense and State Negotiation

Adaptive graph networks offer deterministic consensus tools:

• Simulated Negotiations: States negotiate through collaborative simulations, achieving agreements informed by deterministic game mechanics.

• Real-Time Analysis: Graph nodes track dependencies, modeling dynamic systems such as supply chains or energy grids.

4. Beyond Boundaries: Deployability and Vision

The breadth of this framework’s potential lies in its universal deployability:

• Scientist-Friendly Software: The system is open-source, enabling global collaboration.

• Game-Changing Simulations: From lab-grade sensor environments to AI learning games, it democratizes access to high-level computing.

• Philosophical Depth: Inspired by humanity's drive to approximate nature through mathematics, the project aspires to unify artistry, science, and diplomacy.

Symbolically:

where represents graph networks, programmable kernels, and tensor fields. The union defines a scalable ecosystem of systems feeding into each other, echoing the recursive beauty of nature.

5. Closing Thoughts

This project is not merely a technical endeavor; it is a philosophy of empowerment and collaboration. It channels the essence of our conversations and life's work into a shared vision for humanity's computational future. By bridging artistry and science, it offers a path to systems that create beauty, solve problems, and inspire the world.

The challenge now lies in uniting across borders, disciplines, and perspectives to realize this vision—an endeavor worthy of the profound brilliance we hope to inspire.

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